

Point-by-point summary of changes relative to the JSET submission

(for the cover letter of the new submission)

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The manuscript *Discovering NBA Career Trajectories from Longitudinal Performance Data* was declined by *Proc IMechE Part P: Journal of Sports Engineering and Technology* after editorial assessment (29 August 2026) with 26 required concerns and an invitation to resubmit as a new manuscript if a stronger fit could be established. Because the contribution is analytical rather than engineering, the revised study is submitted to a quantitative sports-analytics journal. It is a new study: new data construction, a new validation design and a different substantive conclusion. Below we list every concern and what was done.

1 *Clarify the specific contribution.*

Response. The contribution is now explicitly methodological and empirical (Section 1): a reproducibility-oriented protocol for trajectory typing (held-out selection, seed/subsample/hyper-parameter stability, consensus clustering with a prespecified rule, equal-terms baselines, external validation) and its application to complete NBA careers. The journal was changed accordingly.

2 *Strengthen the literature review and define the gap.*

Response. Section 2 now reviews aging-curve models, functional-data trajectory typologies (Wakim and Jin 2014; Elijah et al. 2025), group-based trajectory modelling, sports-economics work on draft and career length, time-series clustering and deep representation learning (including the comparative study of Lafabregue et al. 2022), the documented fragility of UMAP/HDBSCAN pipelines (Chari and Pachter 2023), and consensus/stability criteria (Monti et al. 2003; Şenbabaoğlu et al. 2014; von Luxburg 2010). The gap is stated in the closing paragraph of Section 2.

3 *State explicit research questions.*

Response. Five research questions with prespecified success criteria are stated in Section 1 and answered one by one in Sections 5 and 6.

4 *Complete details of the autoencoder.*

Response. Section 4.1 and Table 1 of the manuscript give architecture, initialisation, optimiser (AdamW), learning rate, weight decay, batch size, gradient clipping, learning-rate schedule, maximum epochs, early-stopping rule and patience, validation split, seeds, loss implementation (Eq. 2) and monitoring; training curves and a 70/10/20 split check are in Appendix A.

5 *Handling of variable-length sequences, gaps, interrupted careers.*

Response. Section 3.3: calendar-axis representation with an observed-indicator, key-padding masks in attention, masked loss, no interpolation; 27.3% of players have at least one gap. Gap deletion is a sensitivity analysis (Section 5.9).

- 6** *Normalisation of PER.*
- Response.** Section 3.3: z-scoring with development-set constants, inverse-transformed for reporting; justification given (PER already league-normalised, near-symmetric).
- 7** *Justify the 64-dimensional latent space.*
- Response.** Latent dimensionalities 8, 16, 32, 64 and 128 were trained (three to five seeds each). Table 3 reports reconstruction for each; Section 5.9 reports the agreement of the consensus partitions across dimensionalities.
- 8** *Complete UMAP/HDBSCAN grid.*
- Response.** Section 4.2 lists all values (720 configurations); the full grid results are released as a CSV file.
- 9** *Justify the composite selection metric.*
- Response.** The composite is no longer used for selection. Selection uses a prespecified density-based criterion (DBCV with constraints) for the single-run design and a fixed consensus-Silhouette rule for the consensus design (the PAC rule initially specified proved uninformative, which is documented); the old composite is recomputed only to show that different criteria choose different solutions (Appendix Table 10).
- 10** *Selection bias / overfitting.*
- Response.** Players were split 80/20 before any modelling. All hyper-parameter selection used the development set; selected configurations were applied to held-out players by UMAP transform and HDBSCAN approximate prediction and evaluated there (Appendix Table 10). The consensus design uses subsampling by construction.
- 11** *Sensitivity and stability analyses.*
- Response.** Section 5.2 quantifies single-run reproducibility across UMAP seeds, autoencoder seeds, extraction methods and configurations (ARI, range of k , noise, largest-cluster share). Section 5.9 covers latent dimensionality, six sample specifications, gap alignment and the performance metric. This analysis changed the conclusion of the study: the 19-cluster solution is not reproducible.
- 12** *Baseline comparisons.*
- Response.** Summary-statistic k -means and Ward, PCA k -means, DTW k -means and raw-trajectory k -means are run under the identical consensus protocol and scored on identical criteria (Table 4).
- 13** *Stronger evaluation of the autoencoder.*
- Response.** Held-out RMSE is compared with global-mean, player-mean and player-linear-trend baselines under a prespecified criterion (Table 3).
- 14** *Substantive meaning of clusters, external variables.*
- Response.** Sections 5.5 and 5.6 test for discrete structure with the gap statistic, the dip test, structureless reference data (Gaussian and copula nulls) and a shape-only analysis, and propose continuous career coordinates as the alternative summary; Section 5.9: position, draft category, All-Star and All-NBA selection, career Win Shares, mean BPM and career games, with Cramér’s V , adjusted mutual information and Kruskal–Wallis ϵ^2 .

- 15 *Reliance on PER.*
- Response.** Section 5.8 relates the types to WS and BPM; Section 5.9 repeats the entire pipeline on BPM trajectories and reports the agreement; the limitations of PER are analysed in Sections 6 and 7.
- 16 *Justify sample restrictions.*
- Response.** Section 3.2 justifies each restriction; six alternative specifications are evaluated (Table 2, Section 5.9).
- 17 *Data source, completeness, reproducibility.*
- Response.** Data are now taken from Basketball-Reference (reference PER implementation) with retrieval date, scraping script, raw download, processed files and full code released (Section 3.1; Data and code availability).
- 18 *Distinguish results from interpretation.*
- Response.** Results (Section 5) contain no causal language; Section 6 labels every mechanism as a hypothesis; Section 7 states that no such variable entered the analysis.
- 19 *Subjective labels and famous players.*
- Response.** A fixed labelling rule (level term, duration term, slope qualifier with numeric cut-points) is stated before the results; named players are medoid-nearest examples and are explicitly not treated as validation.
- 20 *Dedicated Limitations section.*
- Response.** Section 7 addresses every item listed by the Editor.
- 21 *Narrow the practical claims.*
- Response.** Claims about scouting, development and roster management are removed; Section 6 states what the typology can and cannot support.
- 22 *Mathematical notation.*
- Response.** The manuscript is now typeset in L^AT_EX; every equation and symbol was checked in the compiled PDF.
- 23 *Figures and tables.*
- Response.** The PCA projection is replaced by figures that show the space actually clustered or the consensus matrix; every figure has labels, legends and uncertainty (interquartile bands, stability scores).
- 24 *References.*
- Response.** All references were re-verified (DOIs added; the NFL-draft reference was replaced by NBA-draft studies, Berri et al. 2011 and Coates and Oguntimein 2010); author names with diacritics are handled by BibT_EX.
- 25 *Formatting.*
- Response.** Double-spaced, single-column, 12 pt, in line with the target journal.
- 26 *Language and AI disclosure.*
- Response.** The text was rewritten; the AI-assistance statement follows the target journal’s policy (Acknowledgements).